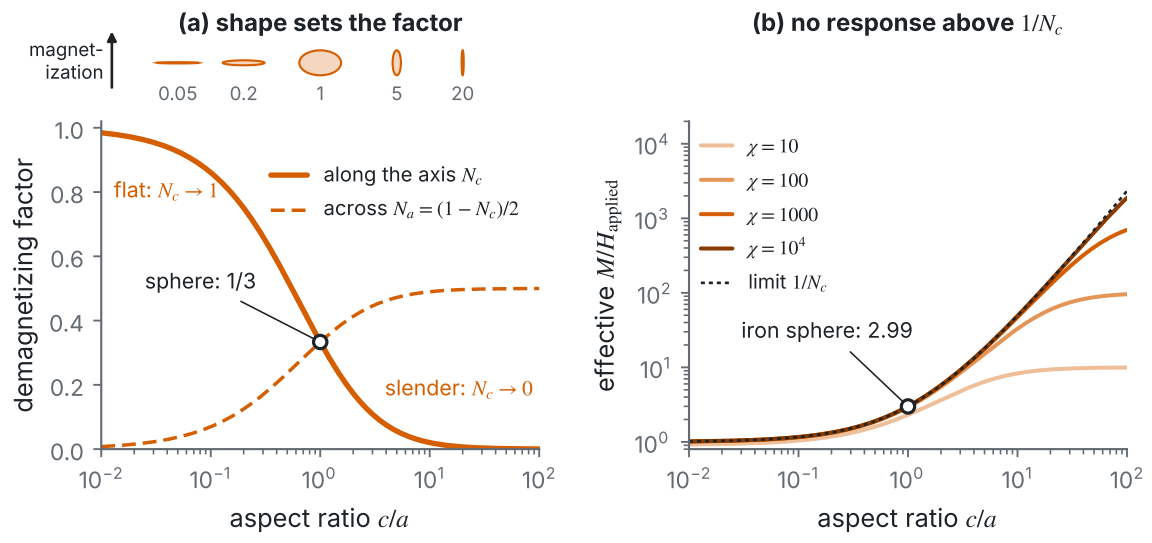
(a) shape sets the factor
magnet-
ization
0.05
0.2
1
5
20
demagnetizing factor
1.0
0.8
0.6
0.4
0.2
0.0
flat: $N_c \to 1$
along the axis N_c
across $N_a = (1 - N_c)/2$
sphere: 1/3
slender: $N_c \to 0$
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
aspect ratio c/a
(b) no response above $1/N_c$
effective M/H_{applied}
10^{4}
10^{3}
10^{2}
10^{1}
10^{0}
$\chi = 10$
$\chi = 100$
$\chi = 1000$
$\chi = 10^4$
limit $1/N_c$
iron sphere: 2.99
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
aspect ratio c/a